Review of Related Literature and Studies

Technical and Theoretical Background

AI ExamGuard: An AI-Assisted Online Examination Proctoring and Academic Integrity Monitoring System

Introduction

This chapter reviews the technical and theoretical foundations underpinning AI ExamGuard, an AI-assisted online examination proctoring system built to detect identity mismatches, unauthorized devices, and behavioral anomalies during remote assessments while remaining auditable and appealable. The technical background surveys the computer-vision and machine-learning techniques behind the system's face-detection, face-recognition, object-detection, head-pose, and risk-scoring components, together with the documented fairness limitations of these techniques. The theoretical background situates the system within academic-integrity, surveillance, and technology-acceptance scholarship, since a proctoring tool is as much a social intervention as a technical one. Together, these two strands motivate the specific design choices — disclosed limitations, an appeals workflow, and a fairness audit — that distinguish this system from conventional black-box commercial proctoring tools.

Technical Background

AI-Based Online Examination Proctoring Systems

The shift to remote and online assessment has driven a large body of work on AI-assisted proctoring, which typically combines webcam-based identity verification, gaze/behavior monitoring, and browser-activity tracking to flag potential academic dishonesty without a live human proctor watching every student continuously. This literature consistently reports that such systems improve scalability and cost relative to live human proctoring, but just as consistently flags privacy, algorithmic bias, and test-taker anxiety as open, unresolved problems rather than solved ones. AI ExamGuard's own architecture — a FastAPI backend coordinating browser-based face and object checks against a React exam-taking client — follows this same general pattern, but was deliberately built to treat the privacy, bias, and anxiety concerns as first-class design requirements (evidence-retention limits, a student appeals workflow, a documented fairness audit) rather than deferred future work, directly responding to the gap this literature repeatedly identifies.

Real-Time Object Detection for Unauthorized-Device Monitoring

Redmon, Divvala, Girshick, and Farhadi (2016) introduced You Only Look Once (YOLO), reframing object detection as a single regression problem over a full image rather than a multi-stage classify-then-localize pipeline, which made real-time detection (45+ FPS on then-current hardware) practical for the first time. Successive YOLO generations, culminating in Ultralytics' YOLOv8 (Jocher, Chaurasia, & Qiu, 2023), retained this single-pass design while improving small-object and occluded-object accuracy — the exact failure mode (a partially hidden phone held low, out of frame center) that matters for exam-room device detection. AI ExamGuard's object-detection service uses a YOLOv8-based general detector alongside a phone-specialist model fine-tuned on real proctoring footage, plus a pose-guided hand-region re-check, specifically to recover the low-confidence, partially-occluded cases this literature identifies as YOLO's persistent weak point.

Lightweight Real-Time Face Detection

Locating a face reliably, at low latency, on commodity webcam hardware without a GPU dependency, is a separate problem from recognizing whose face it is. Wu, Peng, and Yu (2023) address exactly this with YuNet, an anchor-free, sub-100KB-parameter face detector reporting millisecond-level inference and explicit robustness to pose and partial occlusion — properties that matter directly for a browser-based, CPU-only proctoring pipeline that must run a face check every few seconds without becoming the exam-taking experience's bottleneck. AI ExamGuard adopts YuNet (via OpenCV's DNN face module) as its face-localization stage, and the same facial landmarks YuNet already computes for detection are reused downstream for head-pose estimation rather than discarded, directly building on the model's documented landmark output.

Face Recognition for Identity Verification

Once a face is located, identity verification is a distinct recognition problem. Ahonen, Hadid, and Pietikäinen (2006) proposed Local Binary Pattern Histograms (LBPH), which encode a face as a set of local micro-texture histograms rather than holistic pixel intensities, giving the representation documented robustness to the uneven, uncontrolled lighting typical of a home webcam setup. This lighting robustness and low computational cost are why AI ExamGuard trains a per-student LBPH model (via OpenCV's cv2.face module) from a small number of enrollment images rather than requiring a large labeled dataset or cloud inference, at the cost of the lower accuracy headroom the literature also documents relative to modern deep face-recognition embeddings.

Head-Pose Estimation as an Attention/Gaze Proxy

Because true eye-gaze tracking typically needs specialized hardware or tightly controlled camera geometry, online-learning-engagement research instead commonly estimates coarse head orientation from ordinary webcam frames — via OpenCV's solvePnP against a generic 3D face template fitted to a small set of 2D facial landmarks — and treats sustained head-down or off-screen orientation as an attention proxy rather than a literal measurement of visual focus. Closer to the proctoring context specifically, Shih et al. (2024) built an AI-assisted gaze-detection tool at Duolingo to help human proctors triage which recorded exam moments are worth reviewing. This same distinction — pose as a proxy signal, not a ground-truth measurement — is central to how AI ExamGuard frames its own “Prolonged Downward Gaze” feature: head pitch is estimated via solvePnP from YuNet's landmarks at zero added inference cost, but the feature is deliberately disclosed to students as a coarse signal for human review, not as gaze tracking, matching this literature's own caution that head pose is a meaningfully weaker signal than genuine gaze.

Algorithmic Fairness and Demographic Bias in Facial Analysis

A now-foundational finding in this space is Buolamwini and Gebru's (2018) Gender Shades study, which measured commercial gender-classification systems' error rates across a dataset intentionally balanced by skin type and found error rates below 1% for lighter-skinned men but as high as 34.7% for darker-skinned women — evidence that aggregate accuracy figures can conceal large, systematic subgroup disparities. NIST's own Face Recognition Vendor Test Part 3 (Grother, Ngan, & Hanaoka, 2019), evaluating nearly 200 algorithms from around 100 developers, corroborated this at much larger scale, finding demographic-dependent false-positive/false-negative differentials that vary by algorithm and threshold rather than being uniformly present or absent. This literature directly motivates AI ExamGuard's own fairness-audit work: a per-subject/per-batch sensitivity sweep against real proctoring footage, and — because the project's own dataset lacks labeled demographic attributes — a set of disclosed proxy analyses (image-brightness/lighting condition, synthetic camera-quality degradation, and headscarf presence via Meta's FACET benchmark) run as an honest, caveated first step toward the genuine demographic audit this literature shows is necessary, not a substitute for it.

Statistical Risk Scoring and Model Evaluation

Beyond detection and recognition, combining multiple weak behavioral signals (face loss, device detection, tab-switching, prolonged head-down) into a single interpretable risk estimate is itself a standard supervised-learning problem — logistic regression, cross-validation for generalization estimation, and their associated bias-variance tradeoffs are covered in standard statistical-learning references such as James, Witten, Hastie, and Tibshirani (2021). AI ExamGuard's risk service follows this methodology directly: a logistic-regression model fit on real, labeled proctoring windows, with leave-one-subject-out cross-validation — rather than a random split — used specifically to estimate how the score is likely to generalize to a student the model has never seen, the harder and more honest question for a deployed proctoring tool than in-sample accuracy alone.

Theoretical Background

Deterrence Theory and the Fraud Triangle

The premise that observation, or the credible threat of it, reduces dishonest behavior traces to Cressey's (1953) fraud-triangle model, which identifies opportunity, pressure, and rationalization as the three jointly necessary conditions for fraud, and holds that removing any one of them suppresses the behavior. Applied to remote assessment, this predicts that visible, credible monitoring — reducing perceived opportunity and the ease of rationalizing “no one will know” — should measurably reduce dishonesty, a prediction empirically tested by Alguacil, Herranz-Zarzoso, and Perniás (2024), whose randomized field experiment on webcam-monitored versus unmonitored online exams found monitored students scored significantly lower, consistent with monitoring suppressing behavior that had been inflating unmonitored scores. This theory is the direct justification for AI ExamGuard's existence as a deterrent, but it also supplies a caution the system's own design tries to take seriously: deterrence theory says nothing about false positives, and a system that deters by threat of detection carries a distinct due-process obligation when a detection is wrong — the motivation behind building a real student-facing appeals workflow rather than treating every flagged violation as a settled verdict.

Surveillance Theory: The Panopticon and Educational AI

Foucault's (1977) panopticon — a prison architecture in which a single unseen watcher can observe any cell at any moment, so inmates internalize discipline under the mere possibility of being watched — is the standard theoretical lens for continuous monitoring technologies, and has been applied specifically to online proctoring: the one-sidedness of the arrangement, in that the student cannot verify when or whether they are being watched or algorithmically scored, reproduces the panopticon's central mechanism. Dai, Thomas, and Rawolle (2025) extend this analysis to AI-mediated education specifically, arguing that algorithmic surveillance introduces a “post-panoptic” form of disciplinary power in which the system itself, not a human watcher, normalizes behavior toward efficiency and compliance, often at a documented cost to student autonomy and trust. This theoretical framing is why AI ExamGuard treats disclosure as a design requirement rather than a legal formality: every monitored signal, including weaker, proxy-based ones like prolonged downward gaze, is listed to the student before the exam begins, on the reasoning that an undisclosed panopticon is a strictly worse position for the student than a disclosed one, even though disclosure alone does not resolve the power asymmetry the theory describes.

Technology Acceptance Model (TAM)

Davis's (1989) Technology Acceptance Model explains adoption behavior through two constructs — perceived usefulness and perceived ease of use — and remains the dominant framework for studying whether users actually trust and adopt a given system rather than merely tolerate it. Jiang, Goh, Chen, Liu, and Yang (2023) extended TAM specifically to online-proctoring acceptance during COVID-19, and found that social influence and social presence, rather than usefulness alone, were the strongest predictors of whether students accepted being proctored — meaning a proctoring tool's perceived legitimacy depends heavily on institutional framing and social context, not just on the tool's technical accuracy. This motivates AI ExamGuard's positioning choice: since raw detection accuracy is not, by this literature, sufficient for acceptance, the system is deliberately framed to instructors and students as a transparent, explainable, and appealable alternative to opaque commercial proctoring — an attempt to influence the social-presence and trust variables TAM identifies as decisive, not just the underlying model accuracy.

Disability, Accessibility, and Fairness in Surveillance-Based Assessment

A growing qualitative literature documents that proctoring surveillance and disability accommodation are frequently in direct tension: Kwapisz, Ackerman, Nguyen, and Rajivan (2025) interviewed students with disability accommodations and found real, reported anxiety that ordinary accommodation behavior, such as looking away, stepping away, or atypical posture and movement, would itself be misread by the system as suspicious, increasing cognitive load during an already high-stakes exam. This is the direct rationale for AI ExamGuard's accommodation pathway — server-side flags that suppress specific checks (face verification, object detection) entirely for students with a documented need, enforced in the detection service itself rather than only hidden in the frontend — a design response to exactly the failure mode this literature reports, rather than a generic accessibility checkbox.

Related Studies

The literature reviewed above is predominantly foreign. This section reviews local, Philippine-context studies that ground the same technical and theoretical concerns in the population and institutions AI ExamGuard is actually built for.

Academic Dishonesty and Online-Learning Behavior in Philippine Higher Education

Local Philippine research confirms that the deterrence dynamics summarized in the Theoretical Background are not merely imported theory. Aguilar (2021) documented the prevalence of academic dishonesty among Filipino 21st-century learners, and Beruin (2022) surveyed the specific factors driving it during the Philippines' pandemic-era shift to fully online instruction, finding that reduced direct observation was a repeatedly cited enabling condition rather than a side detail. Perez, Zapanta, Heradura, and Napicol (2025) sharpened this into a quantitative account, surveying 562 Filipino undergraduates and finding that demographic factors and, most relevant here, students' own attitude toward cheating were the strongest predictors of self-reported academic cheating in online learning specifically, not distance-learning circumstances alone. Together these three studies establish that the deterrence and fraud-triangle mechanisms discussed earlier operate concretely in the Philippine undergraduate population AI ExamGuard is built for, supporting the premise that a locally-deployed monitoring tool addresses a real, locally-documented problem rather than an imported one.

Technology Acceptance of E-Learning and Learning Management Systems in the Philippines

The Technology Acceptance Model literature reviewed earlier has its own local track record. Garcia (2017) extended TAM with Philippine-specific predictors, including internet connectivity experience and social-media influence, to explain Filipino college students' acceptance of learning management systems, finding these context-specific factors mattered alongside the classic usefulness/ease-of-use constructs. Liday and Agapito (2020) applied an extended TAM specifically to LMS adoption by faculty at a Philippine state university, again finding acceptance was not explained by system quality alone. Both studies reinforce the conclusion that a monitoring or e-learning tool's local acceptance depends on more than technical accuracy — a caution AI ExamGuard's own positioning as a transparent, appealable system is built to take seriously in a Philippine higher-education deployment context specifically, not as a generic TAM footnote.

Local Deployment of Face-Recognition-Based Monitoring in Philippine Educational Institutions

Closest in kind to AI ExamGuard's own face-verification component are two independent local deployments. Grefaldo and Bausa (2025) built a face-recognition attendance-monitoring system for Pilar National Comprehensive High School, reporting concrete gains — eliminated proxy attendance, reduced manual-tracking error — while documenting the data-privacy and access-control obligations (encrypted storage, role-based access) that come with deploying biometric monitoring in a real Philippine school. Valenzuela, Sandigan, Villar, Litang, Obligado, and Masungsong (2025), publishing in Caraga State University's peer-reviewed *Advances in Engineering and Information Sciences*, independently built a comparable facial-recognition computer-laboratory attendance system, formally evaluated against the ISO 9126 software-quality model and rating strongly on functionality, reliability, and usability, while flagging the same recognition-accuracy-under-varying-lighting and privacy concerns as open challenges rather than solved ones. That two separate Philippine institutions building independently arrived at the same encryption/access-control/lighting-sensitivity concerns is strong local corroboration that face-recognition-based monitoring is both technically deployable and institutionally acceptable in Philippine education specifically when paired with real privacy safeguards — and the closest available local precedent to AI ExamGuard's own enrollment/verification pipeline and its retention and audit-logging controls, which arrived at the same privacy-by-design pairing.

Synthesis

Read together, the technical and theoretical literature above converge on the same conclusion from two different directions: an AI proctoring system's real risk is not merely whether the model detects phones and faces accurately, but what happens when it is wrong, for whom it is more often wrong, and whether the person being watched has any way to contest that. The technical literature shows that vision models carry measurable, non-uniform error across subgroups even at reasonable aggregate accuracy, and that honest generalization estimates require deliberately harder evaluation, subject-held-out rather than merely record-held-out, than a single train/test split. The theoretical literature shows that deterrence justifies monitoring in principle but says nothing about wrongful flags, that surveillance without disclosure or recourse reproduces a specific, well-documented power asymmetry, that acceptance depends on perceived legitimacy as much as accuracy, and that accommodation and surveillance can actively conflict unless accommodation is designed in rather than bolted on. The local studies reviewed above confirm these are not abstract, imported concerns: Philippine undergraduates report cheating along the same attitude/opportunity lines deterrence theory predicts, Philippine LMS-adoption research shows the same legitimacy-over-accuracy pattern TAM predicts, and a real Philippine high-school deployment independently arrived at the same privacy-by-design safeguards this project builds in. AI ExamGuard's concrete departures from a conventional proctoring build — a disclosed-limitations pre-exam notice, a per-subject fairness audit, leave-one-subject-out risk-model validation, a real appeals workflow, and server-enforced accommodation flags — are each a direct response to one of these findings, which is the basis for positioning the system, in this thesis, as an honestly-evaluated and transparently-limited alternative to black-box commercial proctoring rather than a like-for-like clone of one.

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